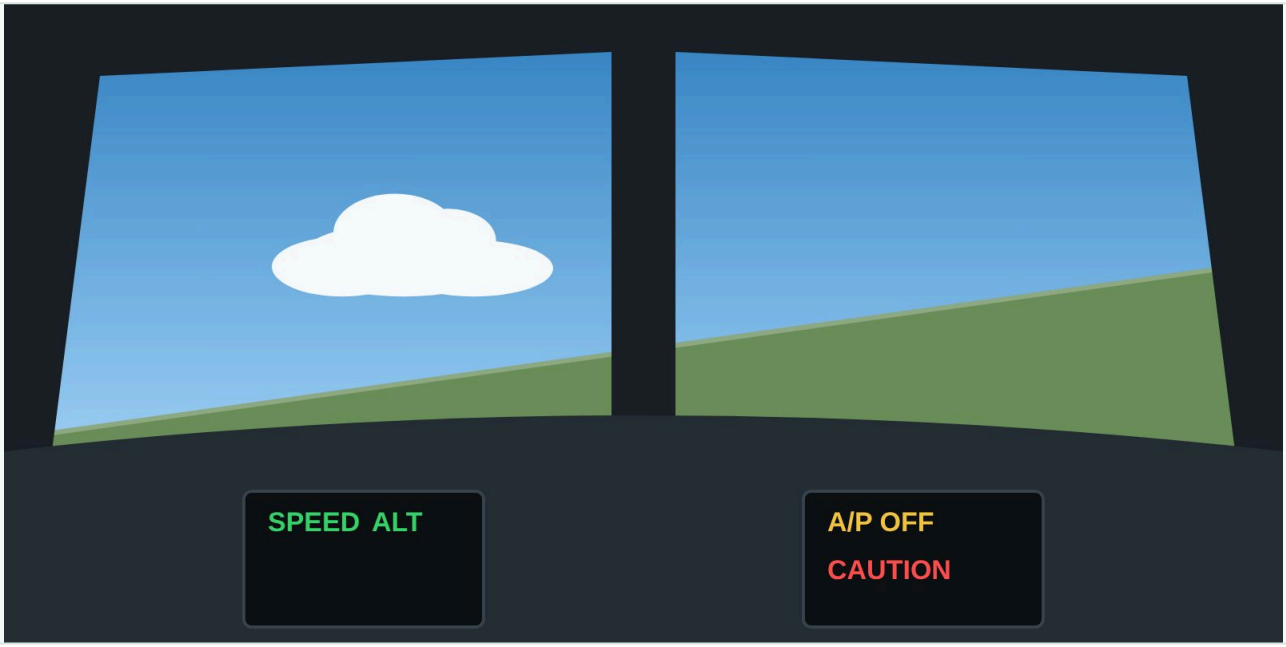
MODULE 4 · THREATS AND HUMAN FACTORS

Lesson 13: Automation, Flight Control Laws and System Failures

Lesson 13 of 24 · about 8 min

LEARNING OBJECTIVES

- Describe how automation can mask or create a developing upset.
- Explain the effects of autopilot disconnect, mode confusion and degraded flight control laws.
- Apply automation management principles to upset prevention.



Automation can mask a developing upset — until it disconnects.

Advantages and limits of automation

Automation flies precisely and reduces workload, which helps prevent many upsets. But it can also:

- **mask** a developing problem — the autopilot holds the aircraft against an asymmetry, ice accretion or mis-trim until it reaches its limit and disconnects, handing the pilot an aircraft that is already out of trim;
- **create** an upset — an unexpected mode, a faulty sensor driving the autothrust or a flight control system, or a disconnect at a critical moment;
- reduce the pilot's awareness of the aircraft's energy state and the manual flying skill needed to recover.

Mode awareness

- The **flight mode annunciator (FMA)** shows what the automation is actually doing. Read it, announce changes, and cross-check.

- Watch for modes where the autothrust is **not** controlling speed — such as idle-thrust descent modes, "hold" modes, or after the autothrust has disconnected.
- If the automation does not do what you expect, **reduce the level of automation** or fly manually — do not spend time reprogramming at low altitude.

Autopilot disconnect

An unexpected disconnect, with its aural warning, is a classic trigger for startle. Be ready for an out-of-trim condition. The pilot flying takes control smoothly, announces it, and stabilises the attitude and speed before doing anything else.

Fly-by-wire control laws

- In **normal law**, envelope protections typically limit bank, pitch, load factor, angle of attack and overspeed.
- Failures — such as unreliable air data or certain system faults — can cause the system to revert to an **alternate or direct law**, in which some or all protections are lost. Handling may also change.
- In a degraded law, the aircraft **can be stalled**. Stall warning and recovery become the pilot's responsibility.

Other system-induced threats

- Engine failure or asymmetric thrust at low speed — yaw and roll that the autopilot may initially hide.
- Flight control malfunctions — jammed or runaway trim, spoiler or rudder problems. Know the memory items.
- Instrument failures — compare the attitude indicators with each other and with the standby instrument.

> **PRINCIPLE**
> Use the level of automation that best suits the situation — and always be able to fly the aircraft manually.

KEY TAKEAWAYS

- Automation can mask a developing upset until it disconnects — expect an out-of-trim aircraft.
- Read and announce the FMA; reduce automation if it does not do what you expect.
- In degraded fly-by-wire laws, protections may be lost and the aircraft can stall.

KNOWLEDGE CHECK

The autopilot disconnects unexpectedly in cruise. What should the pilot flying do first?

A. Re-engage the autopilot immediately

B. Take control smoothly, announce it, and stabilise attitude and speed

C. Check the circuit breakers

D. Call the cabin crew

Answer: B. Fly the aircraft first: take control, announce it, and stabilise the attitude and speed, expecting a possible out-of-trim condition.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.